



Communication And Internet Technology: Mobile IP With IPv6

Mobile IPv6 is a protocol developed as a sub-set of Internet Protocol version 6 (IPv6) to support mobile connections. It allows mobile devices to move from one network to another and still maintain existing connections



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Mobile Internet Protocol (IP) provides mobility support on the existing IP infrastructure. It is needed without any modifications for existing routers, applications or fixed end-hosts. How can then mobile IP change with IPv6?

Mobile IPv6 allows an IPv6 node to be mobile like an IPv4 node. Mobile IPv6, unlike mobile IPv4, supports additional support for mobility, route optimisation and security. IPv6 allows Internet nodes to associate a mobile node's (MN's) home address with its care of address (COA) and send packets directly to COA (and not via home address).

MN obtains COA, eliminating the need for foreign agents. Mobile IPv6 uses route optimisation by which MN sends its current COA to the correspondent node. When it knows about MN's current COA, a correspondent node can deliver packets directly to MN's home address without any assistance from home agent.

Since mobile IPv6 does not make use of foreign agents, it uses correspondent node and binding. Binding refers to an association of home address with COA of MN.

Correspondent node is a peer node with which MN communicates. It may either be mobile or stationary. Its functional operation is given below.

- MN gets registration to foreign network (FN) and gets COA as in mobile IPv4.
- MN sends binding update to home agent.
- Home agent uses proxy neighbour discovery to record MN in home network (HN).
- IP datagram addressed to MN is

encapsulated in IPv6 to IPv6 tunnel. Then, the encapsulated packet is sent to COA of MN.

- IP datagram from MN makes use of the same tunnel in reverse.
- IP datagram from MN is sent to correspondent node directly.
- MN sends binding update to correspondent node.
- Mobility support in IPv6 retains the ideas of HN, home agent and use of encapsulation to deliver packets from HN to MN's current point of attachment, like in IPv4.
- Foreign agents are not required to support mobility in IPv6. IPv6 improves routing efficiency, enhances mobility support and simplifies procedures.

IP telephony and VoIP

The Internet was established primarily for transporting traditional data. Traditional data differs from other communication services, namely, voice and video, in many respects. From customers' point of view, a single network for all services (voice, video and data) sounds appealing.

Single infrastructure, single access is always affordable and convenient. The natural question that arose then was, "Why not use the Internet to carry all services of voice, video and data?"

Solutions for this came in the forms of Internet Protocol telephony (IP telephony)/Voice over Internet Protocol (VoIP) and Internet Protocol TV (IPTV). These solutions address the technical challenges of carrying real-time and jitter-free voice and video over the Internet.

There are several advantages for transmitting VoIP, including: